Evaluate yourself - 6

a) Calculate pH of 10^{-8}M H_2SO_4

B) Calculate the concentration of hydrogen ion in moles per litre of a solution whose pH is 5.4

c) Calculate the pH of an aqueous solution obtained by mixing 50ml of 0.2 M HCl with 50ml 0.1 M NaOH

8.5 Ionisation of weak acids

We have already learnt that weak acids are partially dissociated in water and there is an equilibrium between the undissociated acid and its dissociated ions.

Consider the ionisation of a weak monobasic acid HA in water.

$$HA + H_2O \rightleftharpoons H_3O^+ + A^-$$

Applying law of chemical equilibrium, the equilibrium constant K_c is given by the expression

$$K_C = \frac{[H_3O^+][A^-]}{[HA][H_2O]} \quad .....(8.9)$$

The square brackets, as usual, represent the concentrations of the respective species in moles per litre.

In dilute solutions, water is present in large excess and hence, its concentration may be taken as constant say K. Further H_3O^+ indicates that hydrogen ion is hydrated, for simplicity it may be replaced by H^+. The above equation may then be written as,

$$K_C = \frac{[H^+][A^-]}{[HA] \times K} \quad .....(8.10)$$

The product of the two constants K_C and K gives another constant. Let it be K_a

$$K_a = \frac{[H^+][A^-]}{[HA]} \quad .....(8.11)$$

The constant K_a is called dissociation constant of the acid. Like other equilibrium constants, K_a also varies only with temperature.

Similarly, for a weak base, the dissociation constant can be written as below.

$$K_b = \frac{[B^+][OH^-]}{[BOH]} \quad ....(8.12)$$

8.5.1 Ostwald's dilution law

Ostwald's dilution law relates the dissociation constant of the weak acid (K_a) with its degree of dissociation (α) and the concentration (c). Degree of dissociation (α) is the fraction of the total number of moles of a substance that dissociates at equilibrium.

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